

Local and Global Inequality Rankings under Social Sampling

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Abstract

Survey evidence shows that beliefs about income inequality and own income rank often differ systematically from population statistics. Social sampling provides a natural starting point: people learn about society from comparison sets that are local rather than representative. A recurring feature of subjective rank data is a middle-class bias, with reported positions compressed toward the centre of the income distribution. We use this evidence to motivate rank-symmetric comparison windows centred on the observer and ask what inequality such samples contain. We show that the curvature of the log income profile determines whether inequality in a fixed-width rank window rises or falls with rank, with the main result holding in the Lorenz sense. We then derive a finite-sample approximation for the local Gini and apply the framework to standard income distributions. Finally, we show that local sampling can reverse inequality rankings across societies. For lognormal and Pareto incomes, the local ranking changes at a unique rank cutoff, and global and local rankings can disagree for a majority of rank-local observers. The results isolate a distributional channel through which the same locally centred sampling rule can generate heterogeneous inequality signals.

Keywords: Social sampling; inequality measurement; local Gini; Lorenz order; quantile functions; inequality perceptions

JEL classification: D31; D63; D83

1 Introduction

People have imperfect knowledge of the income distribution. Survey respondents misperceive the level and shape of inequality, and their beliefs about their own position in the income distribution are systematically compressed relative to objective ranks (Page and Goldstein, 2016; Gimpelson and Treisman, 2018; Knell and Stix, 2020). These patterns are relevant because beliefs about inequality and social position enter judgments about redistribution, fairness, and economic opportunity. Yet the informational basis of these beliefs is necessarily local: few individuals observe anything close to a representative sample of the population.

Social-sampling models provide a simple way to formalise this informational constraint. Individuals draw inferences about society from people they encounter in their

immediate social environment, and those environments are themselves structured by homophily, geography, work, family, and other forms of social proximity (Galesic et al., 2012, 2018). Consequently, an apparently biased judgment about society can be consistent with reasonable inference from a biased social sample. In the inequality context, network-based models make the same point: local comparison sets can generate systematic differences between the distribution people observe and the distribution that exists in the population (Schulz et al., 2022; Mamunuru et al., 2025).

This paper combines the social-sampling perspective with a second empirical regularity: the middle-class bias in subjective social position. Poorer respondents tend to place themselves above their objective rank, while richer respondents tend to place themselves below it, so that reported positions are concentrated around the middle of the distribution (Evans and Kelley, 2004; Knell and Stix, 2020). We use this pattern to motivate a tractable benchmark for local comparison sets. An individual at rank r observes the same number of nearby ranks below and above r . The resulting sample is rank-symmetric around the observer. Social sampling and the middle-class bias motivate a perception-formation algorithm in which the observer is approximately central in the income distribution that is locally visible.

Once rank symmetry is imposed, a new question arises. What inequality does an observer see? A symmetric window contains the same number of lower and higher ranks at every position in the distribution, but it need not contain the same amount of income dispersion. The mapping from rank to income may be almost linear in one part of the distribution and strongly curved in another. A fixed movement in rank can therefore correspond to very different proportional income changes at different ranks. The same sampling rule can expose a low-income, middle-income, and high-income observer to systematically different local inequality.

Our first result characterises this variation without imposing a particular parametric income distribution. For equal-width rank windows, the curvature of log income determines the Lorenz ordering of windows located at different ranks. If log income is convex in rank, moving the window upward makes the local distribution more unequal in the Lorenz sense; if log income is concave, local inequality falls with rank. The continuous version of the result is expressed in terms of the curvature of the log quantile function. Because the result is a Lorenz ordering, it applies to any inequality measure that respects the Lorenz criterion, with the Gini coefficient as its most prominent instance.

This characterisation is useful because familiar income distributions imply qualitatively different local inequality profiles. A linear quantile function produces declining local inequality. A Pareto distribution produces increasing local inequality. Exponential and lognormal distributions are non-monotone because the curvature of their log quantile functions changes sign. In the lognormal case, local inequality falls through the lower half of the distribution and rises through the upper half. Thus a common, rank-symmetric sampling rule does not imply a common inequality signal. Observers can all be locally central while seeing different amounts of inequality.

The Lorenz-ordering result establishes direction but not magnitude. We therefore derive a finite-sample approximation for the Gini coefficient within a symmetric rank window. To first order, the local Gini is governed by the proportional slope of the quantile function: local inequality is high where a small movement in rank corresponds to a large proportional movement in income. We provide the discrete-window coefficient and an explicit remainder bound, which allows the approximation to be assessed for finite populations and finite social samples. Numerical simulations demonstrate that

the approximation is accurate for the narrow windows relevant to the social-sampling interpretation.

The second part of the paper asks whether local sampling preserves inequality comparisons across societies. This need not be the case. Two countries may have an unambiguous ordering according to their population Ginis while rank-local observers receive the opposite ordering. We connect this problem to the star-shaped, or quantile-ratio, order (Shaked and Shanthikumar, 2007; Okamoto, 2022; Chan, 2024). A monotone quantile ratio guarantees that a common rank-sampling rule preserves the corresponding Lorenz and Gini ordering. When the quantile ratio changes direction, local order reversals become possible.

A lognormal–Pareto comparison gives a particularly transparent application. Both distributions are standard benchmarks for income data, yet their local inequality profiles differ sharply. We show that their local ranking changes at a unique rank cutoff. Moreover, there is a nonempty parameter region in which the Pareto distribution exhibits higher inequality measured by the Gini coefficient, while a majority of sufficiently local rank windows rank the lognormal distribution as more unequal. The disagreement follows systematically from the shape of the two income distributions and the location of the observer.

The paper contributes to several related strands of literature. It extends work using Pen’s parade to relate the shape of the income-rank schedule to inequality (Milanovic, 1997; Ogwang, 2007; Mussard et al., 2011; Sadefo Kamdem, 2012) by studying the inequality contained in moving segments of the parade rather than only the population distribution. It is related to work on truncation and inequality (Behdani et al., 2020), but focuses on equal-width rank windows and their ordering as the window moves through the distribution. More generally, the paper belongs to the theoretical literature on inequality measurement and distributional orderings, including axiomatic work on inequality measures (Bosmans and Öztürk, 2026) and applications of Lorenz crossings to economic behaviour (Ibragimov et al., 2018). Finally, it complements the literature on social sampling and inequality perceptions by isolating the distributional content of a locally centred sample before specifying how an individual maps that sample into a belief about society.

The remainder of the paper introduces rank-symmetric social sampling and the local inequality measures used in the analysis. It then establishes the moving-window Lorenz ordering and derives a finite-sample approximation for the local Gini. The following sections apply the results to standard income distributions and to cross-society inequality rankings. The final section discusses how the local inequality signals can enter models of inequality perception.

2 Rank-symmetric social sampling

2.1 Comparison sets

Let a population of size n have strictly positive incomes ordered by rank,

$$0 < x_1 \leq x_2 \leq \cdots \leq x_n.$$

An individual at rank r is assumed to observe a local comparison set containing w ranks below and w ranks above their own rank,

$$\mathcal{W}_r = \{r - w, \dots, r, \dots, r + w\}, \tag{1}$$

for interior ranks $r = w + 1, \dots, n - w$. The window contains $m = 2w + 1$ observations.

Rank symmetry means symmetry in the ordered population, not symmetry in income levels. The individual occupies the central rank in the sample by construction, but the income distances on the two sides need not be equal. This distinction is central to the analysis: the sampling rule is fixed across ranks, while the population's income geometry is allowed to vary.

For the ordering results, it is useful to work with any contiguous window of a given length. Define

$$W_i = (x_i, x_{i+1}, \dots, x_{i+m-1}), \quad i = 1, \dots, n - m + 1.$$

The ego-centred sample in (1) is the special case $i = r - w$.

2.2 Local Lorenz curves and Gini coefficients

Let

$$S_i = \sum_{j=0}^{m-1} x_{i+j}$$

denote the total income in window W_i . Its Lorenz ordinates at the support points k/m are

$$L_i\left(\frac{k}{m}\right) = \frac{\sum_{j=0}^{k-1} x_{i+j}}{S_i}, \quad k = 1, \dots, m. \quad (2)$$

The associated local Gini is simply the Gini coefficient computed within the comparison set. It measures the inequality contained in the incomes locally observed around a given rank.

To connect the discrete sample to a population distribution, write

$$p = \frac{r}{n+1}, \quad h = \frac{1}{n+1},$$

and let Q denote the population quantile function. Then the income at rank $r + k$ is

$$x_{r+k} = Q(p + kh), \quad k = -w, \dots, w.$$

For the symmetric window, the local Gini can be written exactly as

$$G_{n,w}(p) = \frac{2 \sum_{k=-w}^w k Q(p + kh)}{(2w + 1) \sum_{k=-w}^w Q(p + kh)}. \quad (3)$$

This expression is the basic object of the analysis. It keeps the rank structure of the social sample explicit and will allow both exact ordering results and a controlled approximation for the level of local inequality.

3 Quantile geometry and moving-window inequality

3.1 Finite-population result

Theorem 1 (Moving-window Lorenz ordering). *Let $x_1, \dots, x_n > 0$ be ordered incomes and define the proportional income increase between adjacent ranks as*

$$\gamma_j = \frac{x_{j+1} - x_j}{x_j}.$$

Suppose that γ_j is nondecreasing in j on the relevant range. Equivalently, $\log x_j$ is discretely convex. Fix a window length m . Then for any two admissible windows W_i and W_{i+d} with $d > 0$,

$$L_{i+d}(u) \leq L_i(u) \quad (4)$$

at every Lorenz ordinate $u = k/m$, $k = 1, \dots, m-1$. Hence

$$G(W_{i+d}) \geq G(W_i). \quad (5)$$

If γ_j is strictly increasing on the ranks connecting the two windows, the inequalities are strict at all interior Lorenz ordinates. If γ_j is nonincreasing, equivalently if $\log x_j$ is discretely concave, all inequalities are reversed.

Proof. Fix i and $d > 0$. For $j = 0, \dots, m-1$, define

$$a_j = x_{i+j}, \quad b_j = x_{i+d+j}, \quad R_j = \frac{b_j}{a_j}.$$

The ratio R_j measures how much larger the income at a given position in the later window is relative to the corresponding position in the earlier window. For adjacent positions within the two windows,

$$\frac{R_{j+1}}{R_j} = \frac{x_{i+d+j+1}/x_{i+d+j}}{x_{i+j+1}/x_{i+j}} = \frac{1 + \gamma_{i+d+j}}{1 + \gamma_{i+j}}.$$

Because γ_j is nondecreasing and $d > 0$, this ratio is at least one. Hence R_j is nondecreasing in j : when the window is shifted upward, incomes at its upper positions increase proportionally at least as much as incomes at its lower positions.

For $k = 1, \dots, m$, define

$$A_k = \sum_{j=0}^{k-1} a_j, \quad B_k = \sum_{j=0}^{k-1} b_j = \sum_{j=0}^{k-1} a_j R_j.$$

The ratio B_k/A_k is the a_j -weighted average of R_j over the first k positions. Since R_j is nondecreasing, this prefix average cannot exceed the corresponding weighted average over all m positions:

$$\frac{B_k}{A_k} \leq \frac{B_m}{A_m}.$$

Rearranging gives

$$\frac{B_k}{B_m} \leq \frac{A_k}{A_m},$$

which is (4). The Gini inequality follows from Lorenz ordering. The case of nonincreasing proportional income changes reverses the monotonicity of R_j and therefore the Lorenz ordering. $\square$

Remark 2. Theorem 1 is exact. It requires neither a large population nor a small window nor symmetry around an ego rank. What matters is that the compared rank windows have the same length.

3.2 Continuous quantile analogue

Let $Q : (0, 1) \rightarrow (0, \infty)$ be an increasing quantile function. For a centre $p \in (\delta, 1 - \delta)$ and half-width $\delta > 0$, consider a local sample that is uniform in population rank over the interval $[p - \delta, p + \delta]$. Formally, let

$$P_{p,\delta} \sim \text{Unif}[p - \delta, p + \delta]$$

and define the corresponding local income variable as

$$Y_{p,\delta} = Q(P_{p,\delta}).$$

Because Q is increasing, the quantile function of this local income distribution is

$$q_{p,\delta}(u) = Q(p - \delta + 2\delta u), \quad u \in [0, 1]. \quad (6)$$

Thus $q_{p,\delta}$ maps each within-window quantile u to the corresponding income in the rank interval $[p - \delta, p + \delta]$. Where Q is differentiable, convexity of $\log Q$ is equivalent to the proportional rank slope $Q'(p)/Q(p)$ being nondecreasing.

Theorem 3 (Log-quantile curvature and local Lorenz order). *Let $g(p) = \log Q(p)$. If g is convex on an interval containing two equal-width rank windows centred at $p_1 < p_2$, then*

$$L_{p_2,\delta}(u) \leq L_{p_1,\delta}(u) \quad \text{for all } u \in [0, 1], \quad (7)$$

and therefore

$$G_{p_2,\delta} \geq G_{p_1,\delta}.$$

If g is concave on that interval, the inequalities are reversed.

Proof. Write

$$q_1(u) = q_{p_1,\delta}(u), \quad q_2(u) = q_{p_2,\delta}(u), \quad R(u) = \frac{q_2(u)}{q_1(u)}.$$

Let $d = p_2 - p_1 > 0$ and write

$$x(u) = p_1 - \delta + 2\delta u.$$

Then

$$\log R(u) = g(x(u) + d) - g(x(u)).$$

For a convex function g , the fixed-length increment

$$g(x + d) - g(x)$$

is nondecreasing in x . Since $x(u)$ is increasing in u , $R(u)$ is therefore nondecreasing. This argument does not require differentiability of g . Since $q_2(u) = q_1(u)R(u)$, the q_1 -weighted average of R over $[0, u]$ is no larger than its q_1 -weighted average over $[0, 1]$:

$$\frac{\int_0^u q_2(v) dv}{\int_0^u q_1(v) dv} \leq \frac{\int_0^1 q_2(v) dv}{\int_0^1 q_1(v) dv}.$$

Rearrangement yields (7). The concave case is symmetric. □

Corollary 4 (Rank profile of local inequality). *On any rank interval on which $\log Q$ is convex, equal-width local rank distributions become weakly more unequal as the window moves upward. On any interval on which $\log Q$ is concave, they become weakly less unequal.*

The result has a simple interpretation. Suppose proportional income gaps become larger as rank increases. Then moving a fixed-width social window upward magnifies the relative distances between incomes inside the sample, so the later window is more unequal. If proportional gaps become smaller with rank, the opposite occurs. Because the comparison is in the Lorenz sense, this conclusion does not depend on choosing the Gini as the particular measure of inequality.

Figure 1 illustrates the result for a lognormal distribution. Below the median, moving the comparison set upward from $p = 0.2$ to $p = 0.4$ raises the local Lorenz curve and reduces inequality. Above the median, moving it from $p = 0.6$ to $p = 0.8$ lowers the local Lorenz curve and increases inequality. The figure plots differences between Lorenz curves so that the reversal in ordering is visible despite the small size of the local windows.

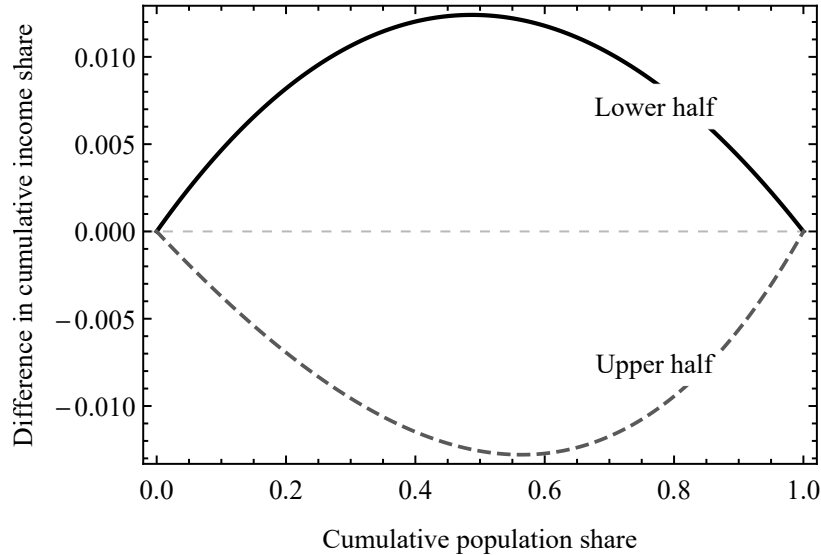


Fig. 1 Finite-sample illustration of the moving-window Lorenz ordering for a lognormal distribution with $\sigma = 1$, $n = 5000$, and $w = 250$. The solid line is the local Lorenz curve at centre $p = 0.4$ minus that at $p = 0.2$; the dashed line is the curve at $p = 0.8$ minus that at $p = 0.6$. A positive difference means that the later window is less unequal, while a negative difference means that it is more unequal.

4 The level of local inequality

The moving-window theorem determines whether local inequality rises or falls with rank, but it does not say how large local inequality is. For the social-sampling application, the magnitude matters as well: two ranks may be ordered in the same direction but differ only negligibly, or they may receive sharply different inequality signals. This section derives a small-window approximation for the exact local Gini in (3).

Let $m = 2w + 1$ and define

$$A_2 = \sum_{k=-w}^w k^2 = \frac{w(w+1)(2w+1)}{3},$$

and

$$A_4 = \sum_{k=-w}^w k^4 = \frac{w(w+1)(2w+1)(3w^2+3w-1)}{15}.$$

Theorem 5 (Finite-sample local-Gini expansion). *Let Q be three times continuously differentiable on $[p - wh, p + wh]$ and suppose*

$$Q(u) \geq q_0 > 0$$

throughout that interval. Let

$$M_j = \sup_{u \in [p-wh, p+wh]} |Q^{(j)}(u)|, \quad j = 1, 2, 3.$$

Then the exact local Gini in (3) satisfies

$$G_{n,w}(p) = \frac{c_w}{3(n+1)} \frac{Q'(p)}{Q(p)} + R_{n,w}(p), \quad (8)$$

where

$$c_w = \frac{2w(w+1)}{2w+1}, \quad (9)$$

and

$$|R_{n,w}(p)| \leq h^3 \left[\frac{M_3 A_4}{3m^2 q_0} + \frac{M_1 M_2 A_2^2}{m^3 q_0^2} \right], \quad h = \frac{1}{n+1}. \quad (10)$$

Consequently, under uniform local bounds on the derivative ratios and with $wh \rightarrow 0$,

$$G_{n,w}(p) = \frac{c_w}{3(n+1)} (\log Q)'(p) + O((wh)^3). \quad (11)$$

Whenever the leading term is bounded away from zero, the corresponding relative approximation error is $O((wh)^2)$.

Proof. Let

$$N = \sum_{k=-w}^w kQ(p+kh), \quad D = \sum_{k=-w}^w Q(p+kh).$$

Equation (3) gives

$$G_{n,w}(p) = \frac{2N}{mD}.$$

Pairing the terms for k and $-k$ and applying Taylor's theorem gives

$$N = hQ'(p)A_2 + E_N, \quad |E_N| \leq \frac{M_3 h^3 A_4}{6}.$$

Similarly,

$$D = mQ(p) + E_D, \quad |E_D| \leq \frac{M_2 h^2 A_2}{2}.$$

Since $D \geq mq_0$, subtracting

$$\frac{2hQ'(p)A_2}{m^2Q(p)}$$

from $2N/(mD)$ yields

$$\left| G_{n,w}(p) - \frac{2hQ'(p)A_2}{m^2Q(p)} \right| \leq h^3 \left[\frac{M_3A_4}{3m^2q_0} + \frac{M_1M_2A_2^2}{m^3q_0^2} \right].$$

Finally,

$$\frac{2A_2}{m^2} = \frac{2w(w+1)}{3(2w+1)} = \frac{c_w}{3},$$

which gives (8). □

Remark 6 (Interpretation). The leading term in (11) is proportional to $(\log Q)'(p)$. This quantity is the proportional increase in income associated with a marginal increase in rank. For a fixed window width, local inequality is therefore larger where nearby ranks are separated by larger proportional income differences. The approximation turns the geometry behind the Lorenz ordering into a quantitative measure of the inequality signal contained in the local sample.

Remark 7 (Accuracy). The bound in (10) also identifies the cases in which the approximation becomes less accurate. The error grows with the relative window width and with local curvature of the quantile function, and it can become large near parts of the support where derivative ratios are large. If $w \rightarrow \infty$ while $w/(n+1) \rightarrow 0$, then $c_w/w \rightarrow 1$, so the familiar continuous-window coefficient w is recovered. For a fixed small window, however, the discrete coefficient c_w is the appropriate one.

Figure 2 evaluates the approximation in finite samples. For an exponential distribution with $n = 1000$ and $w = 5$, the corrected first-order expression is almost indistinguishable from the exact local Gini, while the continuous-window coefficient produces a visible level error. Panel (b) shows the maximum relative error over interior ranks for the exponential, Pareto, and lognormal cases. The error remains small for narrow windows and rises smoothly as the comparison set becomes wider.

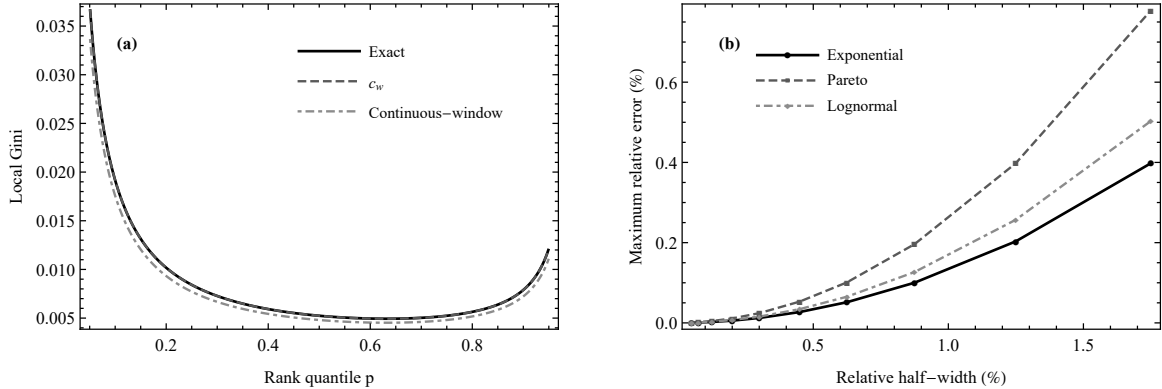


Fig. 2 Accuracy of the local-Gini approximation. Panel (a) compares the exact exponential local Gini with the corrected first-order approximation using c_w and the continuous-window approximation using w , for $n = 1000$ and $w = 5$. Panel (b) reports the maximum relative error of the corrected approximation over $p \in [0.1, 0.9]$ for $n = 4000$ as relative window width changes.

5 Implications for standard income distributions

5.1 Linear quantile function

Let

$$Q(p) = a + bp, \quad b > 0,$$

on a range where $Q(p) > 0$. Then

$$(\log Q)''(p) = -\frac{b^2}{(a + bp)^2} < 0.$$

The log quantile is strictly concave, so equal-width local windows become less unequal as rank increases. The mechanism is transparent: absolute income spacing is constant while the income level used to normalize those gaps rises with rank. Thus the same rank-symmetric comparison rule generates a declining local inequality signal.

For the discrete zero-intercept benchmark $x_r = br$, equation (3) gives

$$G_{n,w}^{\text{lin}}(r) = \frac{2}{3} \frac{w(w+1)}{(2w+1)r}. \quad (12)$$

Appendix A derives exactly the same expression from the covariance representation of the Gini used in the earlier Pen's-parade literature.

5.2 Pareto quantile function

For

$$Q(p) = b(1-p)^{-1/\alpha}, \quad \alpha > 0,$$

$$(\log Q)''(p) = \frac{1}{\alpha(1-p)^2} > 0.$$

The log quantile is strictly convex. Local inequality therefore rises with rank throughout the Pareto region: proportional income gaps widen as the comparison set moves into the upper tail. The small-window approximation makes the rate of this increase explicit:

$$G_{n,w}^{\text{Pareto}}(p) = \frac{c_w}{3(n+1)} \frac{1}{\alpha(1-p)} + O((wh)^3). \quad (13)$$

5.3 Exponential quantile function

For

$$Q(p) = -\frac{1}{\lambda} \log(1-p),$$

write $L(p) = -\log(1-p)$. Then

$$(\log Q)''(p) = \frac{L(p) - 1}{(1-p)^2 L(p)^2}.$$

The log quantile is concave for

$$p < 1 - \frac{1}{e}$$

and convex for

$$p > 1 - \frac{1}{e}.$$

Local inequality therefore falls with rank below the curvature switch and rises above it. The exponential case is thus the simplest example in which a common rank-symmetric sampling rule produces a non-monotone inequality signal. The finite-window approximation is

$$G_{n,w}^{\text{exp}}(p) = \frac{c_w}{3(n+1)} \frac{1}{(1-p)[- \log(1-p)]} + O((wh)^3). \quad (14)$$

The turning point of the leading term remains

$$p^* = 1 - \frac{1}{e},$$

because c_w is a positive multiplicative constant.

5.4 Lognormal quantile function

For

$$Q(p) = \exp\left[\mu + \sigma \Phi^{-1}(p)\right], \quad \sigma > 0,$$

let $z = \Phi^{-1}(p)$. Then

$$(\log Q)'(p) = \frac{\sigma}{\phi(z)}$$

and

$$\text{sign}((\log Q)''(p)) = \text{sign}(z).$$

Hence $\log Q$ is concave below the median and convex above it. The lognormal distribution therefore gives a particularly natural benchmark for the social-sampling application: local

inequality declines as the observer moves through the lower half of the population and increases through the upper half. For small windows,

$$G_{n,w}^{\text{LN}}(p) = \frac{c_w}{3(n+1)} \frac{\sigma}{\phi(\Phi^{-1}(p))} + O((wh)^3), \quad (15)$$

whose leading term has its minimum at $p = 1/2$.

Figure 3 compares the four cases using exact finite-sample local Ginis. The linear and Pareto cases provide opposite monotone benchmarks. The exponential and lognormal cases are non-monotone, but their minima occur at different ranks. The figure therefore makes the central point of the section visible: rank symmetry alone does not determine a universal pattern of local inequality. The pattern depends on the geometry of the underlying income distribution.

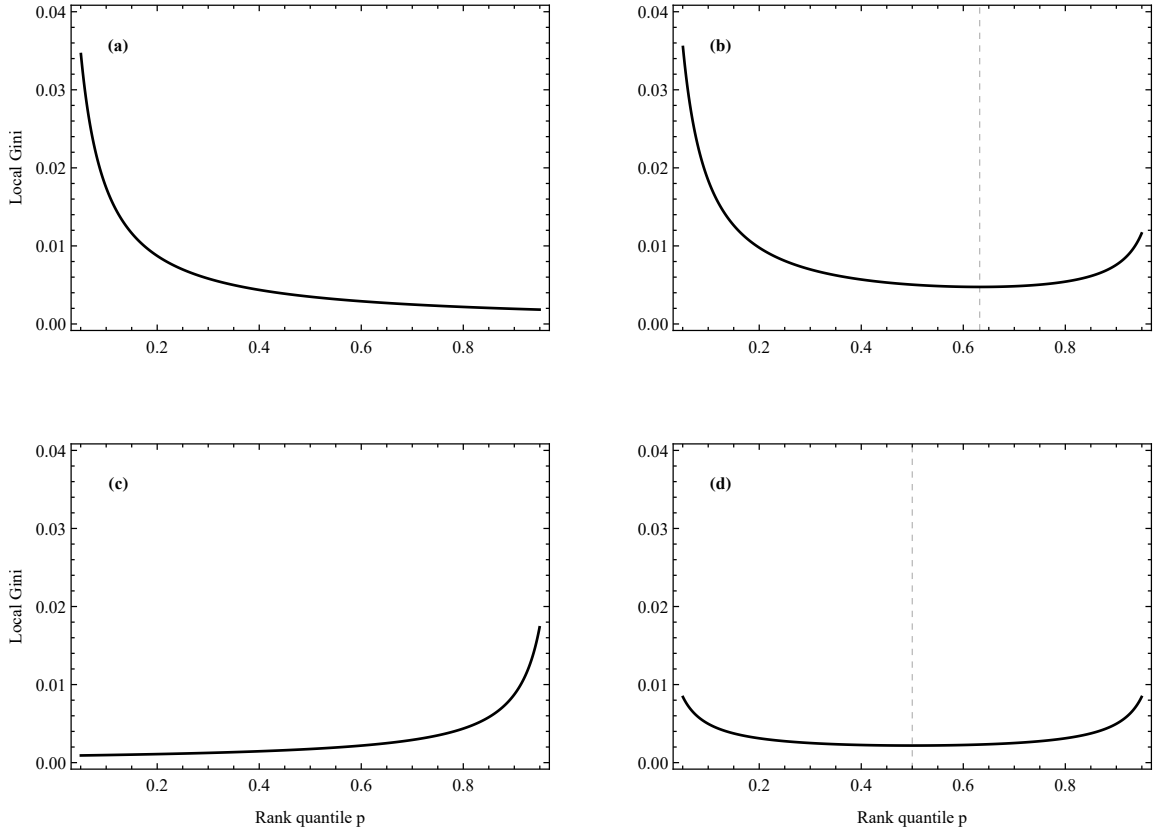


Fig. 3 Exact finite-sample local Gini profiles for (a) linear, (b) exponential, (c) Pareto, and (d) lognormal quantile functions over $p \in [0.05, 0.95]$. The calculations use $n = 2000$, $w = 10$, Pareto tail index $\alpha = 2$, and lognormal scale $\sigma = 0.5$. Vertical reference lines mark the curvature switch in the exponential and lognormal cases.

6 Local versus global inequality rankings

The preceding results concern variation in the inequality signal within one society. A different question arises when individuals compare inequality across societies. A population ranking need not be preserved by local comparison sets: the same ranks can display different relative income spacing even when one society has the larger population Gini.

6.1 Common rank sampling and quantile-ratio order

Consider two societies A and B with positive increasing quantile functions Q_A and Q_B . Let H be any distribution over ranks supported on an interval $I \subset (0, 1)$, and draw a common rank $P \sim H$. The induced local income distributions are

$$Y_A = Q_A(P), \quad Y_B = Q_B(P).$$

Define the quantile ratio

$$R(p) = \frac{Q_A(p)}{Q_B(p)}.$$

If R is nondecreasing, the two distributions satisfy the classical star-shaped order, also described as quantile-ratio dominance (Shaked and Shanthikumar, 2007; Okamoto, 2022; Chan, 2024). This order provides the natural benchmark for asking whether an inequality comparison survives a common rank-sampling rule.

Proposition 8 (Rank-sampling preservation). *If $R(p)$ is nondecreasing on I , then for every common rank-sampling distribution H supported on I , the induced distribution Y_A star-dominates Y_B . Consequently,*

$$Y_A \succeq_L Y_B \quad \text{and} \quad G(Y_A) \geq G(Y_B).$$

If $R(p)$ is nonincreasing on I , all inequalities are reversed.

Proof. Since Q_A and Q_B are increasing, the quantile functions of the induced distributions are

$$Q_{Y_A}(u) = Q_A(H^{-1}(u)), \quad Q_{Y_B}(u) = Q_B(H^{-1}(u)).$$

Their ratio is

$$\frac{Q_{Y_A}(u)}{Q_{Y_B}(u)} = R(H^{-1}(u)).$$

It inherits the monotonicity of R . The stated star, Lorenz and Gini orderings follow. $\square$

Proposition 8 says that a monotone quantile ratio is sufficient for every common rank sample on the interval to preserve the inequality ordering. The next result shows how tightly this condition is connected to local comparisons.

Proposition 9 (Two-rank characterization). *For an interval I , the following statements are equivalent:*

1. $R(p) = Q_A(p)/Q_B(p)$ is nondecreasing on I .
2. For every pair $s < t$ in I , the equally weighted two-rank sample satisfies

$$G(\{Q_A(s), Q_A(t)\}) \geq G(\{Q_B(s), Q_B(t)\}).$$

Proof. For two positive ordered incomes $x < y$,

$$G(\{x, y\}) = \frac{y - x}{2(x + y)},$$

which is strictly increasing in the ratio y/x . Hence the Gini ordering for the pair $s < t$ is equivalent to

$$\frac{Q_A(t)}{Q_A(s)} \geq \frac{Q_B(t)}{Q_B(s)},$$

or equivalently $R(t) \geq R(s)$. $\square$

Corollary 10 (Local order reversals). *Suppose the population Gini satisfies $G_A > G_B$. If R is strictly decreasing on a rank interval I , then every common rank sample supported on I satisfies*

$$G(Y_A) < G(Y_B).$$

Thus a global Gini ranking does not generally survive rank-local sampling.

6.2 Lognormal–Pareto single crossing

The preservation result also identifies where reversals can occur: they require a region in which the quantile ratio moves in the opposite direction. The lognormal–Pareto comparison provides an analytically tractable example with a unique switch in the local ranking.

Let society LN have a lognormal income distribution

$$Q_{\text{LN}}(p) = \exp[\mu + \sigma\Phi^{-1}(p)], \quad \sigma > 0,$$

and let society P have a Pareto distribution

$$Q_{\text{P}}(p) = b(1 - p)^{-1/\alpha}, \quad \alpha > 1.$$

Scale parameters μ and b do not affect the relevant Gini rankings.

Let

$$z = \Phi^{-1}(p)$$

and define the standard normal hazard rate

$$\mathcal{H}(z) = \frac{\phi(z)}{1 - \Phi(z)}.$$

Theorem 11 (Lognormal–Pareto local single crossing). *There is a unique rank cutoff*

$$p^* = \Phi(\mathcal{H}^{-1}(\alpha\sigma)) \tag{16}$$

such that the quantile ratio $Q_{\text{LN}}(p)/Q_{\text{P}}(p)$ is strictly increasing for $p < p^$ and strictly decreasing for $p > p^*$. Consequently:*

1. *every common rank sample supported entirely below p^* satisfies*

$$G_{\text{LN}}^{\text{local}} \geq G_{\text{P}}^{\text{local}};$$

2. *every common rank sample supported entirely above p^* satisfies*

$$G_{\text{LN}}^{\text{local}} \leq G_{\text{P}}^{\text{local}}.$$

Proof. Differentiating the log quantile ratio gives

$$\frac{d}{dp} \log \frac{Q_{\text{LN}}(p)}{Q_{\text{P}}(p)} = \frac{\sigma}{\phi(z)} - \frac{1}{\alpha(1 - p)}.$$

The derivative is positive exactly when

$$\alpha\sigma > \frac{\phi(z)}{1 - \Phi(z)} = \mathcal{H}(z).$$

The normal hazard rate is strictly increasing from zero to infinity, so there is a unique solution $\mathcal{H}(z^*) = \alpha\sigma$. Proposition 8 gives the local Gini rankings on either side. $\square$

The population Ginis are

$$G_{\text{LN}} = 2\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - 1, \quad G_{\text{P}} = \frac{1}{2\alpha - 1}. \quad (17)$$

Define

$$\sigma_G(\alpha) = \sqrt{2}\Phi^{-1}\left(\frac{\alpha}{2\alpha - 1}\right). \quad (18)$$

Then $G_{\text{P}} > G_{\text{LN}}$ exactly when $\sigma < \sigma_G(\alpha)$.

Corollary 12 (Majority order reversal in the local limit). *Define*

$$\sigma_M(\alpha) = \frac{\sqrt{2/\pi}}{\alpha}. \quad (19)$$

For every $\alpha > 1$,

$$\sigma_M(\alpha) < \sigma_G(\alpha).$$

Therefore the interval

$$\boxed{\frac{\sqrt{2/\pi}}{\alpha} < \sigma < \sqrt{2}\Phi^{-1}\left(\frac{\alpha}{2\alpha - 1}\right)} \quad (20)$$

is nonempty. Within this interval the Pareto distribution is globally more unequal, while $p^* > 1/2$. Hence, in the limit of arbitrarily narrow rank neighborhoods, a strict majority of ranks locally order the lognormal distribution as more unequal.

Proof. The condition $p^* > 1/2$ is equivalent to $z^* > 0$, and hence to

$$\alpha\sigma > \mathcal{H}(0) = \sqrt{\frac{2}{\pi}},$$

which gives (19). It remains to show that the lower threshold lies below the global-Gini threshold. At $\sigma = \sigma_M(\alpha)$,

$$\frac{\sigma}{\sqrt{2}} = \frac{1}{\alpha\sqrt{\pi}}.$$

Since the standard normal density is maximized at zero,

$$2\Phi(x) - 1 < \sqrt{\frac{2}{\pi}}x \quad \text{for } x > 0.$$

Thus

$$G_{\text{LN}}(\sigma_M) < \frac{\sqrt{2}}{\pi\alpha} < \frac{1}{2\alpha - 1} = G_{\text{P}},$$

where the second inequality holds for every $\alpha > 1$. Therefore $\sigma_M(\alpha) < \sigma_G(\alpha)$. $\square$

Corollary 12 shows that the disagreement can be systematic: there are parameter values for which the population Gini ranks the Pareto society as more unequal while more than half of sufficiently local rank positions receive the opposite signal. The result is not driven by an isolated tail comparison.

The local-limit argument extends directly to finite-width windows. For a continuous rank window of half-width δ , admissible centres lie in $[\delta, 1 - \delta]$. A window centred at p is guaranteed to rank the lognormal as more unequal whenever $p + \delta < p^*$.

Corollary 13 (Finite-width majority condition). *Let $0 < \delta < 1/2$ and suppose $G_P > G_{LN}$. A strict majority of admissible equal-width rank-window centres are guaranteed to reverse this global ordering whenever*

$$p^* > \frac{1}{2} + \delta, \quad (21)$$

or equivalently

$$\alpha\sigma > \mathcal{H}\left(\Phi^{-1}\left(\frac{1}{2} + \delta\right)\right). \quad (22)$$

For the discrete symmetric window, the natural approximation is $\delta = w/(n+1)$.

Figure 4 illustrates both parts of the argument. For $\alpha = 2$ and $\sigma = 0.5$, the Pareto distribution has the larger population Gini. Yet panel (a) shows that local Ginis rank the lognormal distribution as more unequal over most lower and middle ranks before crossing near the analytical cutoff. Panel (b) shows that this is not a knife-edge example. The shaded parameter region contains all cases in which the global ranking favors greater Pareto inequality while the local cutoff lies above the median, so that a majority of sufficiently local comparison sets reverse the aggregate ranking.

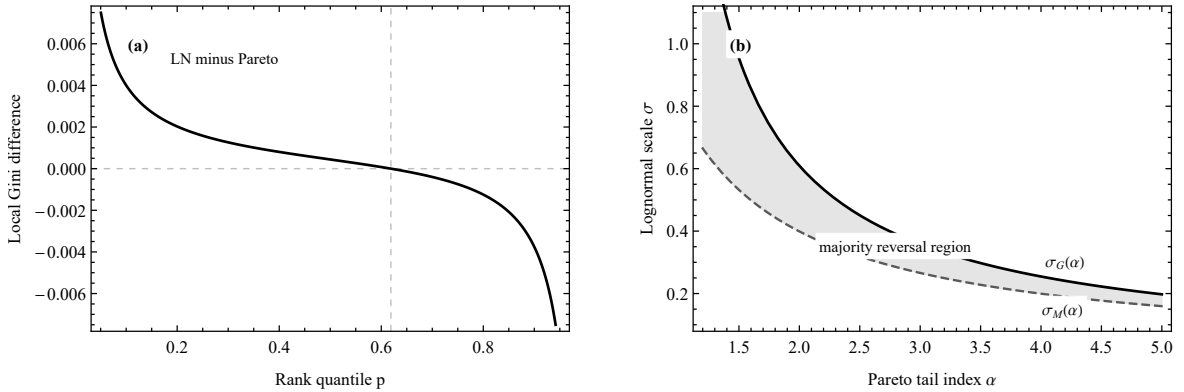


Fig. 4 Lognormal–Pareto order reversal. Panel (a) plots the exact local-Gini difference for $\alpha = 2$, $\sigma = 0.5$, $n = 2000$, and $w = 10$; positive values mean that the lognormal local sample is more unequal. The vertical line marks the analytical cutoff. Panel (b) shades the parameter region in which the Pareto population Gini is larger while a majority of arbitrarily narrow rank neighborhoods rank the lognormal distribution as more unequal.

7 Implications for inequality perceptions

The formal results describe the inequality contained in rank-local comparison sets. To obtain a model of perceived population inequality, one must add a rule that maps the local sample into a belief about society. The social-sampling literature supplies a natural interpretation: individuals use observations from their immediate environment as evidence about a broader population (Galesic et al., 2012, 2018). Recent evidence on inequality perceptions is consistent with such extrapolation. Domènech-Arú (2025), for example, finds that changes in locally observed inequality affect beliefs about national inequality.

Under this interpretation, rank symmetry and the moving-window results play different roles. The middle-class bias motivates the assumption that individuals are approximately central within their local comparison sets. The theorems then characterize the inequality signal generated by those locally centred samples. A person can therefore feel locally middle class and still observe very different inequality from someone at another objective rank. The distinction is important: the model does not explain why subjective rank is compressed toward the centre; it asks what follows for inequality perceptions if local social information has that structure.

Several implications follow. First, rank dependence in perceived inequality need not have a universal sign. With a linear income-rank schedule, higher-ranked observers receive lower local Ginis. In a Pareto tail, the opposite occurs. Lognormal and exponential distributions generate non-monotone patterns. Empirical relationships between objective income rank and perceived inequality may therefore depend on the shape of the income distribution as well as on the composition of social networks.

Second, a low local Gini should not automatically be interpreted as a cognitive bias. It can be an accurate description of inequality inside the observer’s comparison set. A bias relative to population inequality arises only when the observer extrapolates from that local statistic without fully correcting for selection into the sample. This separation between the local signal and the extrapolation rule makes the framework compatible with different behavioral models, ranging from naive extrapolation to partial correction or Bayesian learning from local and aggregate information.

Third, the cross-society results imply that disagreement need not be limited to the level of inequality. Local samples can also reverse the ranking of two societies. If observers compare countries, regions, or periods using socially sampled information, they may disagree with an aggregate Gini ranking even when all observers use the same sampling rule and process their samples in the same way. The lognormal–Pareto example shows that such reversals can affect a majority of rank positions.

The rank-window representation is deliberately parsimonious. Real social environments are neither perfectly rank-local nor perfectly symmetric, and the width of a relevant comparison set is likely to vary across individuals. Network models can endogenize these features (Schulz et al., 2022), while measures of experienced inequality can incorporate network structure directly (Mamunuru et al., 2025). The purpose of the present framework is different. Rank symmetry provides a transparent benchmark, motivated by the empirical middle-class bias, in which the effect of income-distribution geometry can be characterized exactly. This benchmark can subsequently be relaxed to study asymmetric samples, heterogeneous window widths, or mixtures of local and global information.

8 Conclusion

Inequality perceptions are formed under limited and socially structured information. Motivated by social sampling and by the middle-class bias in subjective income rank, this paper studies a simple benchmark in which individuals observe rank-symmetric comparison sets centred on their own position.

The first result shows that the inequality contained in these local samples is governed by the geometry of the income distribution. Log-convex income profiles generate increasing local inequality as the comparison window moves upward, while log-concave profiles generate decreasing local inequality. The result holds in the Lorenz sense and therefore

extends beyond the Gini coefficient. A finite-sample expansion then links the level of the local Gini to the proportional slope of the quantile function and provides an explicit bound on the approximation error.

These general results produce sharply different rank profiles for standard income distributions. Linear and Pareto profiles imply opposite monotone patterns, while exponential and lognormal distributions generate interior minima. Rank symmetry therefore does not homogenize inequality signals across observers. It only fixes the position of the observer inside the local sample.

The same logic applies to comparisons across societies. A global inequality ranking is preserved under common rank sampling when the relevant quantile ratio is monotone, but it can reverse where that condition fails. In the lognormal–Pareto comparison, the reversal occurs at a unique rank cutoff and can affect a majority of local comparison sets even when the population Gini ranking is unambiguous.

The analysis isolates a distributional mechanism rather than a complete theory of belief formation. If individuals extrapolate from local social environments, the results translate into systematic rank dependence in perceived inequality. If they correct for local selection, the same results describe the signal that must be corrected. In either case, the geometry of the income distribution and the structure of the social sample jointly determine the inequality information available to the observer.

A Covariance representation for the linear benchmark

This appendix verifies that the direct window formula and the covariance representation used in the Pen’s-parade literature (Milanovic, 1997; Ogwang, 2007) produce the same local Gini in the linear benchmark.

Consider the symmetric window

$$\{r - w, \dots, r + w\}$$

with $N = 2w + 1$ observations and incomes

$$y_{(r+k)} = b(r + k), \quad k = -w, \dots, w, \quad b > 0.$$

Let $s \in \{1, \dots, N\}$ denote within-window rank. Income is affine in s :

$$Y = b(r - w + s - 1).$$

The within-window mean is

$$\bar{y}_r = \frac{1}{N} \sum_{k=-w}^w b(r + k) = br.$$

For equally weighted observations, the covariance representation of the Gini is

$$G = 2 \frac{\text{Cov}(Y, s)}{N \bar{y}_r}. \quad (23)$$

Since Y is affine in s with slope b ,

$$\text{Cov}(Y, s) = b \text{Var}(s).$$

For the discrete uniform rank variable $s \in \{1, \dots, N\}$,

$$\text{Var}(s) = \frac{N^2 - 1}{12}.$$

Substitution into (23) gives

$$G = 2 \frac{b(N^2 - 1)/12}{Nbr} = \frac{N^2 - 1}{6Nr}.$$

Using $N = 2w + 1$,

$$N^2 - 1 = 4w(w + 1),$$

so

$$G = \frac{2}{3} \frac{w(w + 1)}{(2w + 1)r}, \quad (24)$$

which is identical to (12). Thus the covariance method and the direct moving-window Gini formula are two representations of the same finite-sample object in the linear case. The direct representation in (3) is more convenient for curved quantile functions because it does not require the within-window income–rank correlation to be handled separately.

B Directional windows

For comparison, consider directional windows that contain the same number of observations but place the focal rank at one end of the sample rather than at its centre. In the linear zero-intercept benchmark their local means are

$$\bar{y}_r^{\text{up}} = b(r + w), \quad \bar{y}_r^{\text{down}} = b(r - w),$$

and the covariance derivation above therefore gives

$$G^{\text{up}}(r) = \frac{2}{3} \frac{w(w + 1)}{(2w + 1)(r + w)},$$

and

$$G^{\text{down}}(r) = \frac{2}{3} \frac{w(w + 1)}{(2w + 1)(r - w)}.$$

These formulas show how the linear benchmark changes when rank symmetry is relaxed in a simple directional way.

Supplementary Information

Online Resource 1 (ESM_1.nb) is a Mathematica notebook reproducing the numerical illustrations and numerical checks reported in the manuscript.

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Code availability A Mathematica notebook reproducing the numerical illustrations and numerical checks is provided as Supplementary Information (Online Resource 1).

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